

VANSH GUPTA

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Objective

Pre-final year Electronics and Computer Engineering student with strong foundations in C++, Data Structures, Software Engineering, and System Testing. Experienced in building reliable web applications and validating AI-driven systems through structured testing and analysis.

Education

Thapar Institute of Engineering and Technology

B.Tech, Electronics and Computer Engineering

Patiala, India

2023 – 2027

- CGPA: 9.15 (Top 10% in branch)

Crescent Public School

12th (Non-Medical)

Bilaspur, HP

2022

- Score: 92.2%

Glory Public School

10th Class

Bilaspur, HP

2020

- Score: 98.43%

Skills

Programming: C++, Python, JavaScript
Testing & QA: Test Case Design, Debugging, Unit Testing, Validation, Logs Analysis
Frontend: HTML, CSS, JavaScript, Responsive UI
Tools: Git, GitHub, Postman, VS Code, Figma (UI Basics)
Concepts: DSA, OOP, OS, DBMS, NLP, IoT, Software Testing Life Cycle (STLC)

Projects

Convo-AI – Cognitive Bias Detection and Mitigation System

NLP, Deep Learning, Python, Explainable AI, Web

2025

- Built an AI-based system that detects cognitive biases in real-time user text.
- Designed structured test scenarios to validate model predictions and reduce false positives.
- Fine-tuned a BERT-based model for classification of biased vs unbiased reasoning.
- Developed an Explainable AI module to justify predictions in human-readable form.
- Implemented a web dashboard for real-time analysis with backend logging and monitoring.

GreenSync – Smart Self-Watering Plant System

IoT, ESP-32, Node.js, Express, Sensors, Cloud

2025

- Designed an automated irrigation system using ESP-32 and soil moisture sensors.
- Built a secure Node.js backend with authentication, APIs, and real-time data streaming.
- Conducted functional testing of sensor data reliability and edge cases.
- Developed a web dashboard for live monitoring of temperature, humidity, and soil data.
- Integrated TP4056 battery module for efficient power management.

Achievements

- 3rd Rank in Himachal Pradesh Class 10 Board Exams (2020)
- Top 10% students in branch at TIET
- State-Level Badminton Player